

Report for dataset 20241115_Liverpool_202345_Cell2-Cell2

Dataset 2ca78207-7a44-440b-83c6-23f0c17c246e

Summary

Name	20241115_Liverpool_202345_Cell2-Cell2
Created At	2024-11-17 00:54:26.563
Number of Records	5,864,222
Total Length	86,925,188,574
Movie Name	m84105_241115_134631_s1
ICS Version	13.1.0.221972
Well Sample	20241115_Liverpool_202345_Cell2
Biological Sample	20241115_Liverpool_202345_Cell2

Run Preview

Summary

Estimated HiFi Yield (Gb)	88,827,500,203
HiFi read quality, median	34
HiFi read length, mean	14,632
P1 (%)	None
Time (Hrs)	23
Workflow	Full
Movie Name	m84105_241115_134631_s1_full_preview_23hr
Movie Id	m84105_241115_134631_s1

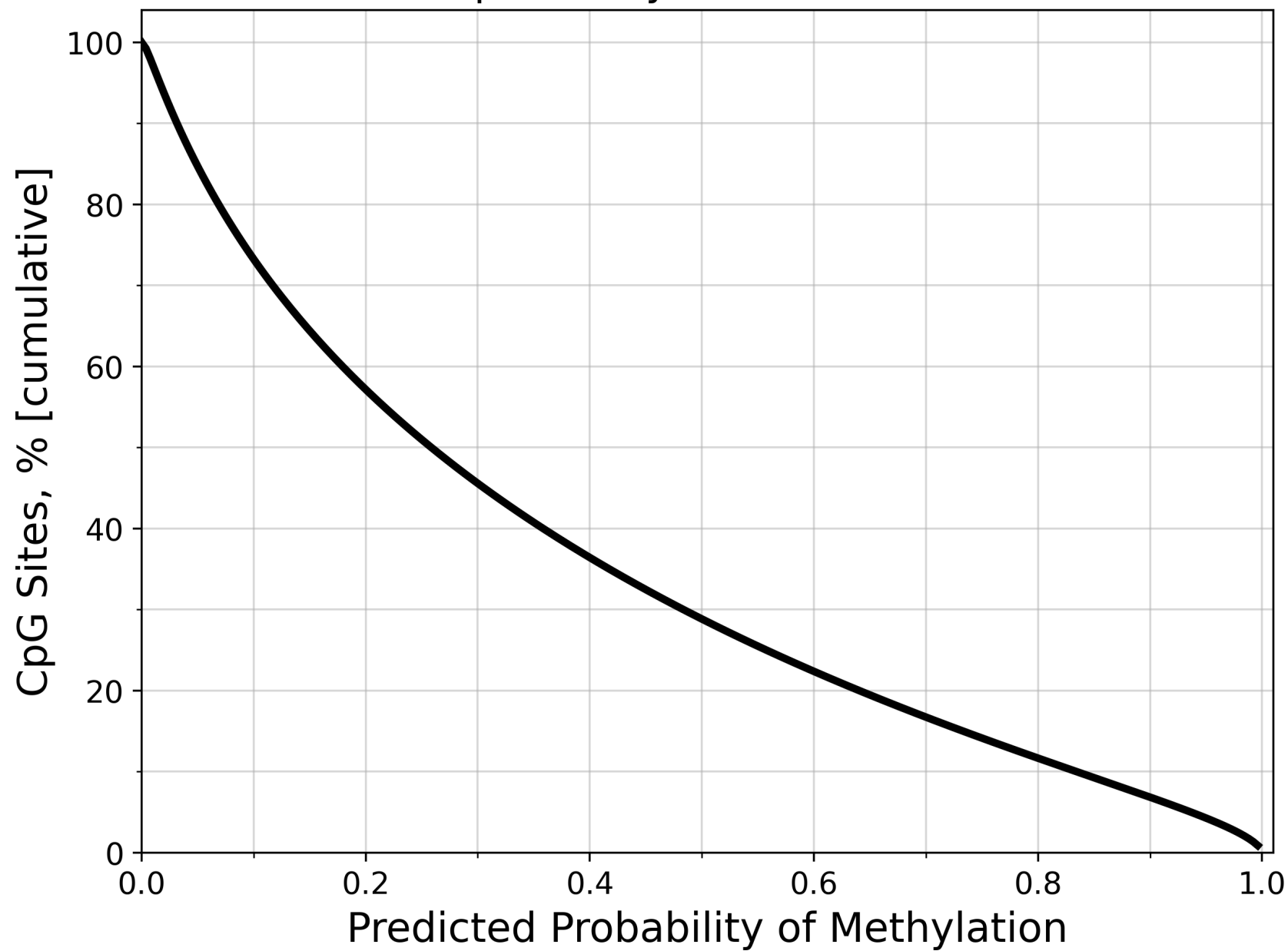
Barcode Preview

Barcode	HiFi reads	HiFi read length (mean, bp)
---------	------------	-----------------------------

5mC CpG Report

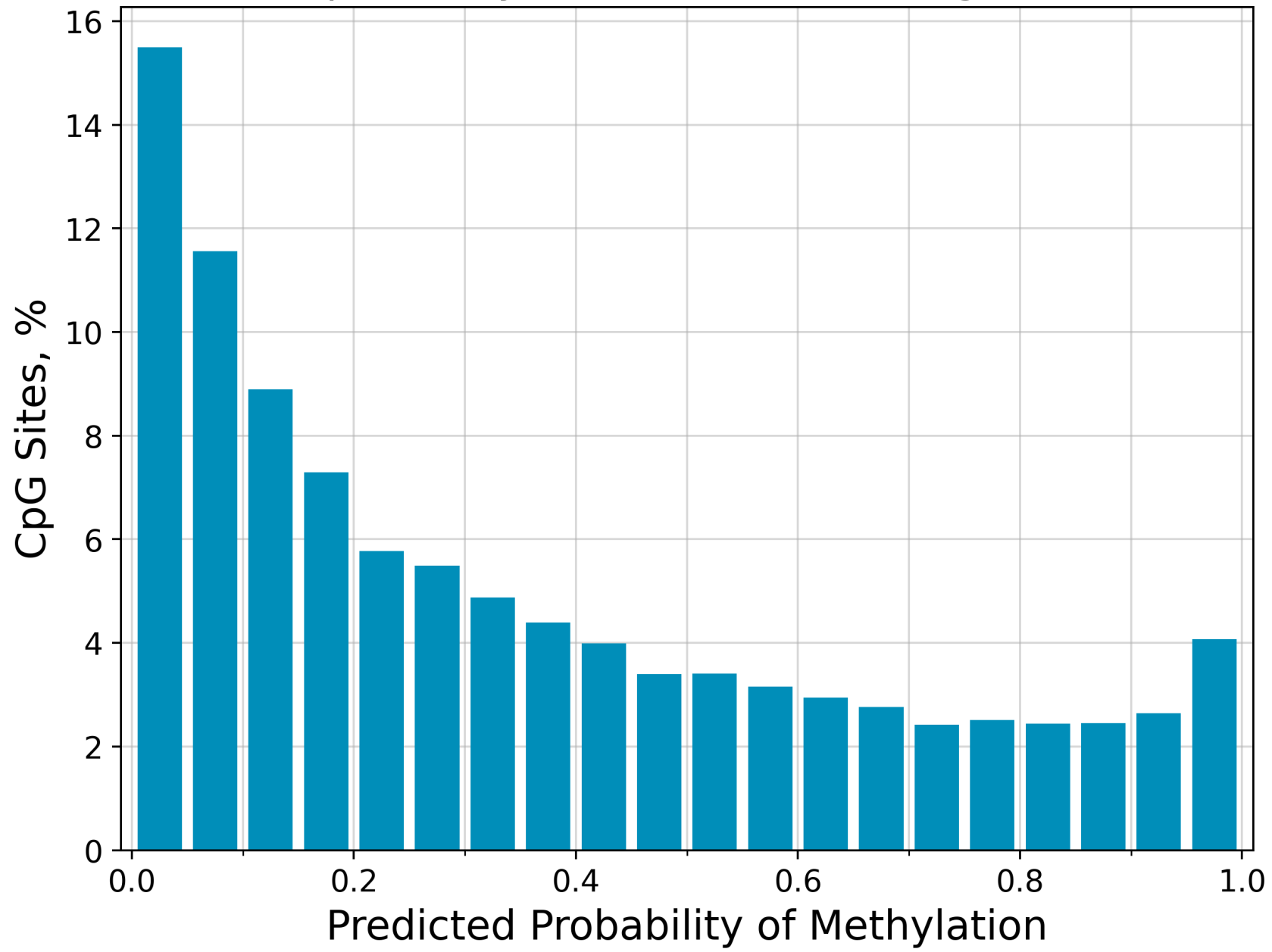
CpG Methylation in Reads

CpG Methylation in Reads



CpG Methylation in Reads (Histogram)

CpG Methylation in Reads (Histogram)



CCS Processing

Summary

ZMWs input	14,842,568
ZMWs pass filters	6,031,693
ZMWs fail filters	8,810,875
ZMWs shortcut filters	0
ZMWs with tandem repeats	72,913
Below SNR threshold	275,284
Median length filter	920,271
Lacking full passes	5,120,773
Heteroduplex insertions	0
Coverage drops	0
Insufficient draft cov	0
Draft too different	0
Draft generation error	0
Draft above --max-length	0
Draft below --min-length	0
Reads failed polishing	4,166
Empty coverage windows	7,206
CCS did not converge	0
CCS adapter concatenation	0
CCS adapter palindrome	2
CCS adapter residue	21
ZMW with full-length subread	2,482,339
ZMW with control failure	694
ZMW with control success	119
CCS below minimum RQ	0
Unknown error	0

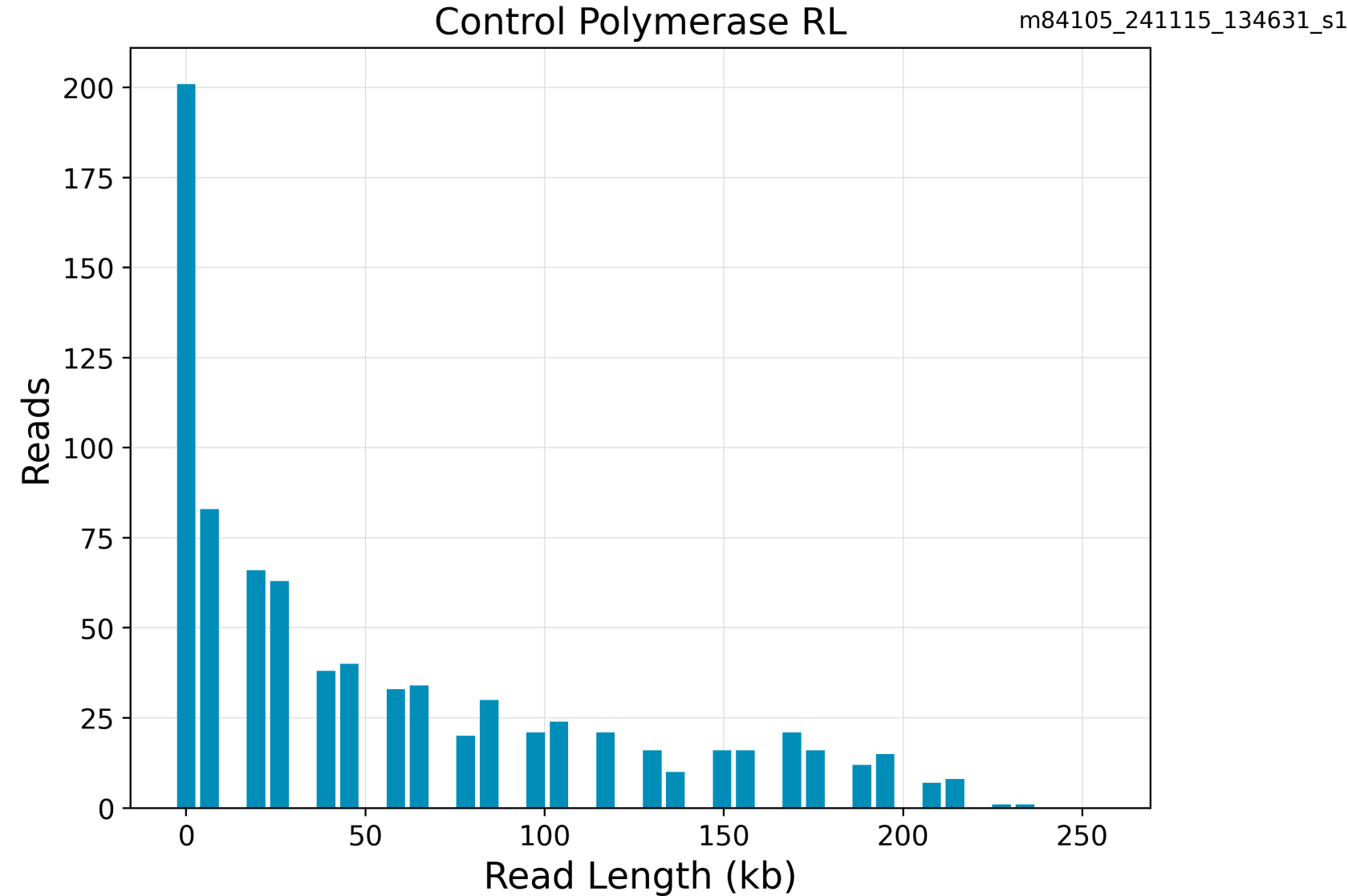
ZMWs missing adapters	87,422
HiFi Reads	5,864,222
HiFi Yield (bp)	86,925,188,574
HiFi Read Length (mean, bp)	14,822
HiFi Read Length (median, bp)	14,123
HiFi Read Length N50 (bp)	14,123
HiFi Read Quality (median)	35
HiFi Number of Passes (mean)	8
<Q20 Reads	167,471
<Q20 Yield (bp)	2,534,695,352
<Q20 Read Length (mean, bp)	15,135
<Q20 Read Length (median, bp)	14,428
<Q20 Read Quality (median)	18
>=Q30 Reads	4,168,637
>=Q30 Yield (bp)	60,070,608,071
>=Q30 Read Length (mean, bp)	14,410
>=Q30 Read Length (median, bp)	13,764
>=Q30 Read Quality (median)	38
Base quality >=Q30 (bp)	81,099,933,232
Base quality >=Q30 (%)	93.30000000000001

Control Report

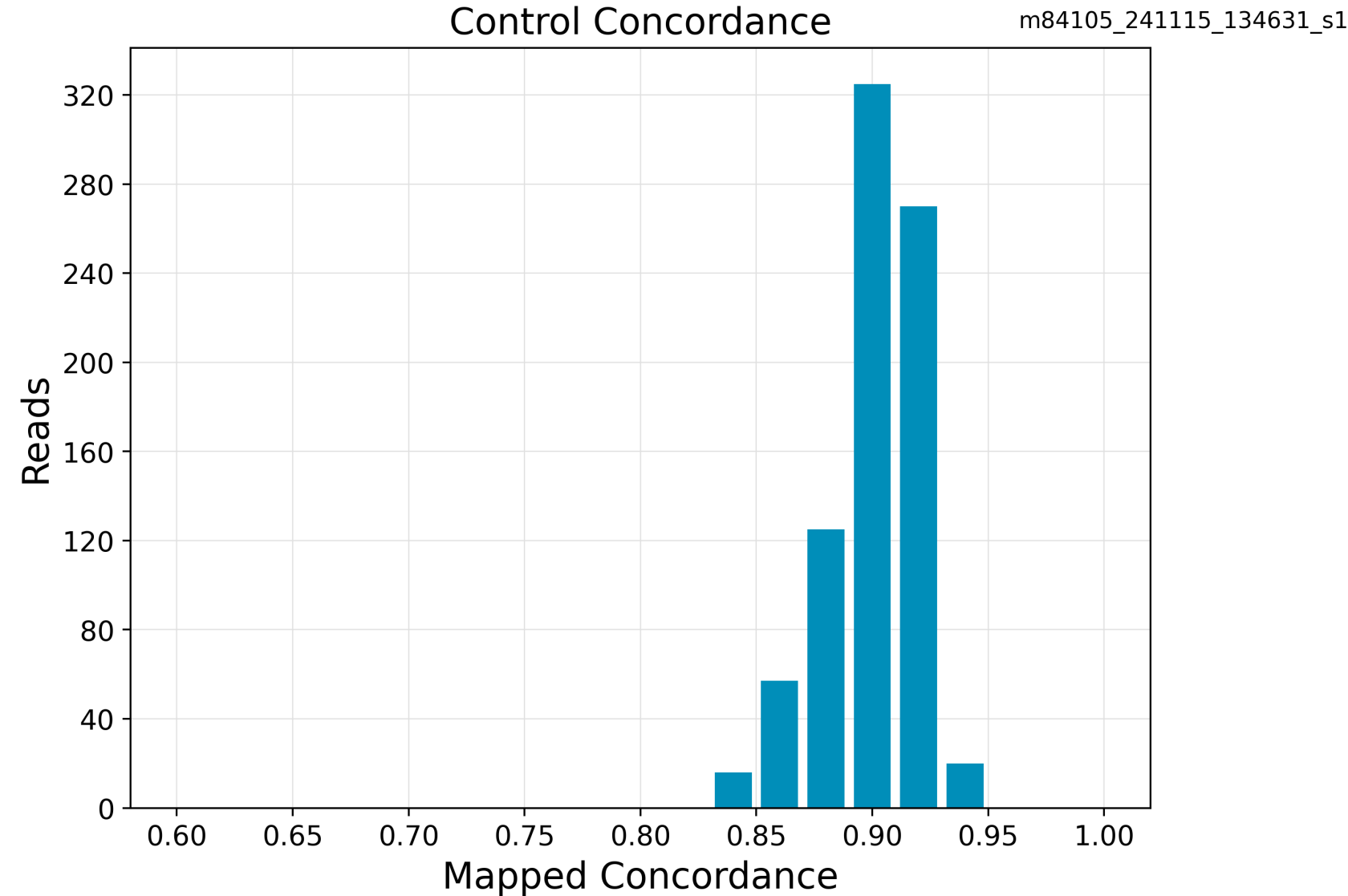
Summary

Number of Control Reads	813
Control Read Length Mean	58,633
Control Read Concordance Mean	0.91
Control Read Concordance Mode	0.91

Control Polymerase RL



Control Concordance



Run Preview

Summary

Estimated HiFi Yield (Gb)	None
HiFi read quality, median	None
HiFi read length, mean	14,161
P1 (%)	54.60 %
Time (Hrs)	4
Workflow	Basic
Movie Name	m84105_241115_134631_s1_basic_preview_4hr
Movie Id	m84105_241115_134631_s1

Barcode Preview

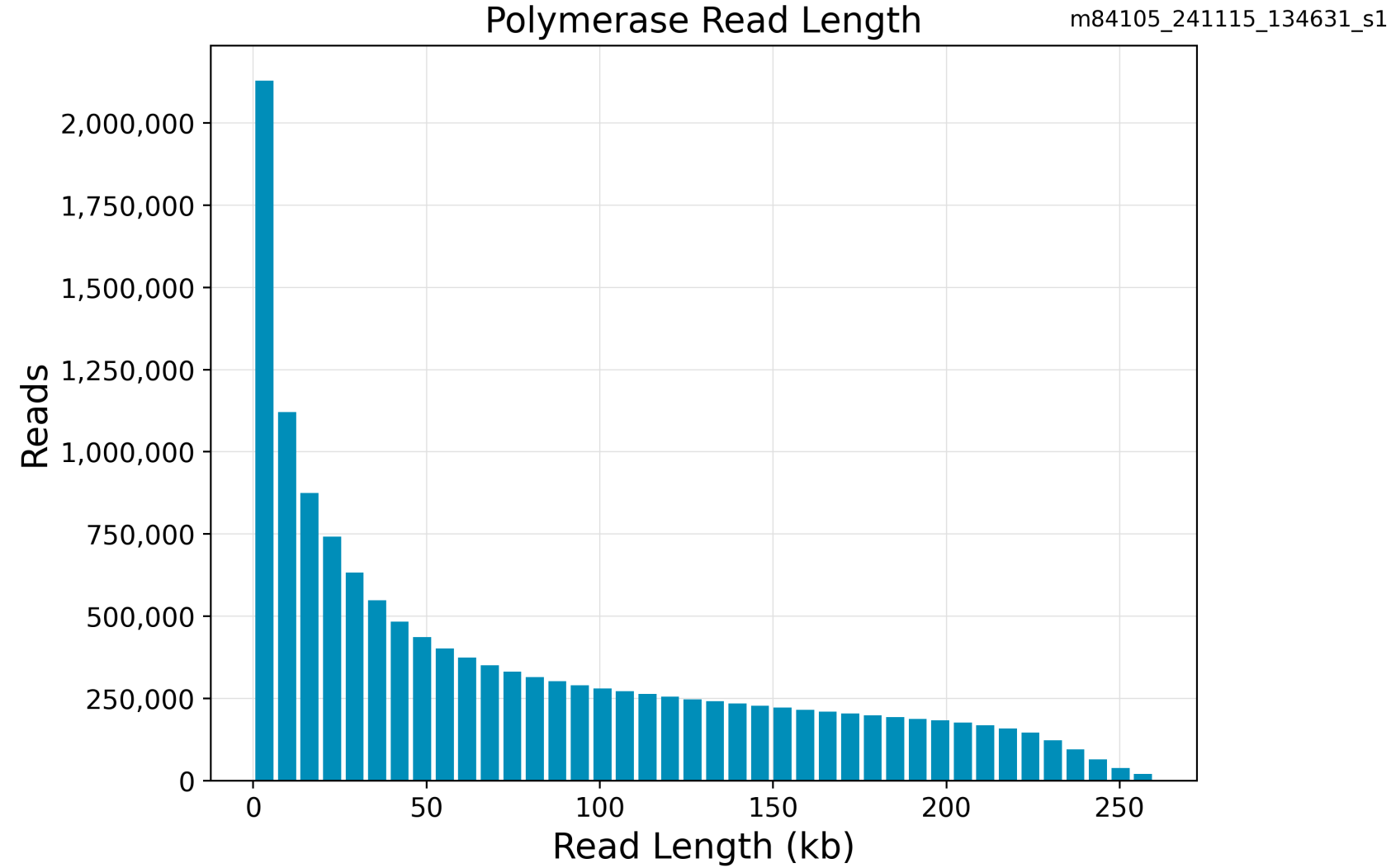
Barcode	HiFi reads	HiFi read length (mean, bp)
---------	------------	-----------------------------

Raw Data Report

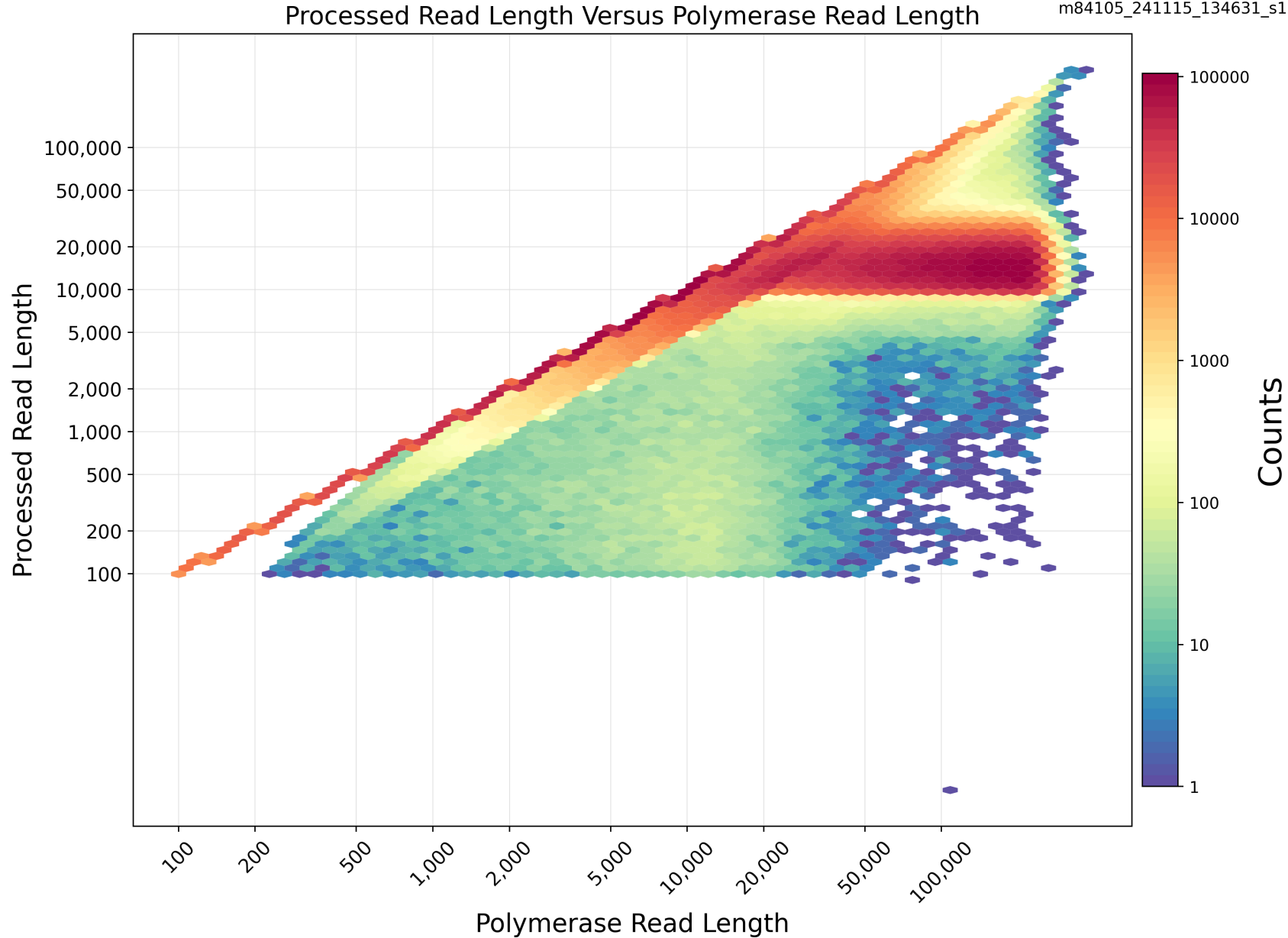
Summary

Polymerase Read Bases	1,034,217,007,465
Polymerase Reads	13,982,689
Polymerase Read Length (mean)	73.96 kb
Polymerase read length (N50)	144.75 kb
Polymerase read length longest subread length (mean)	15.55 kb
Polymerase read length longest subread length (N50)	17.75 kb
Unique Molecular Yield	205,840,760,832

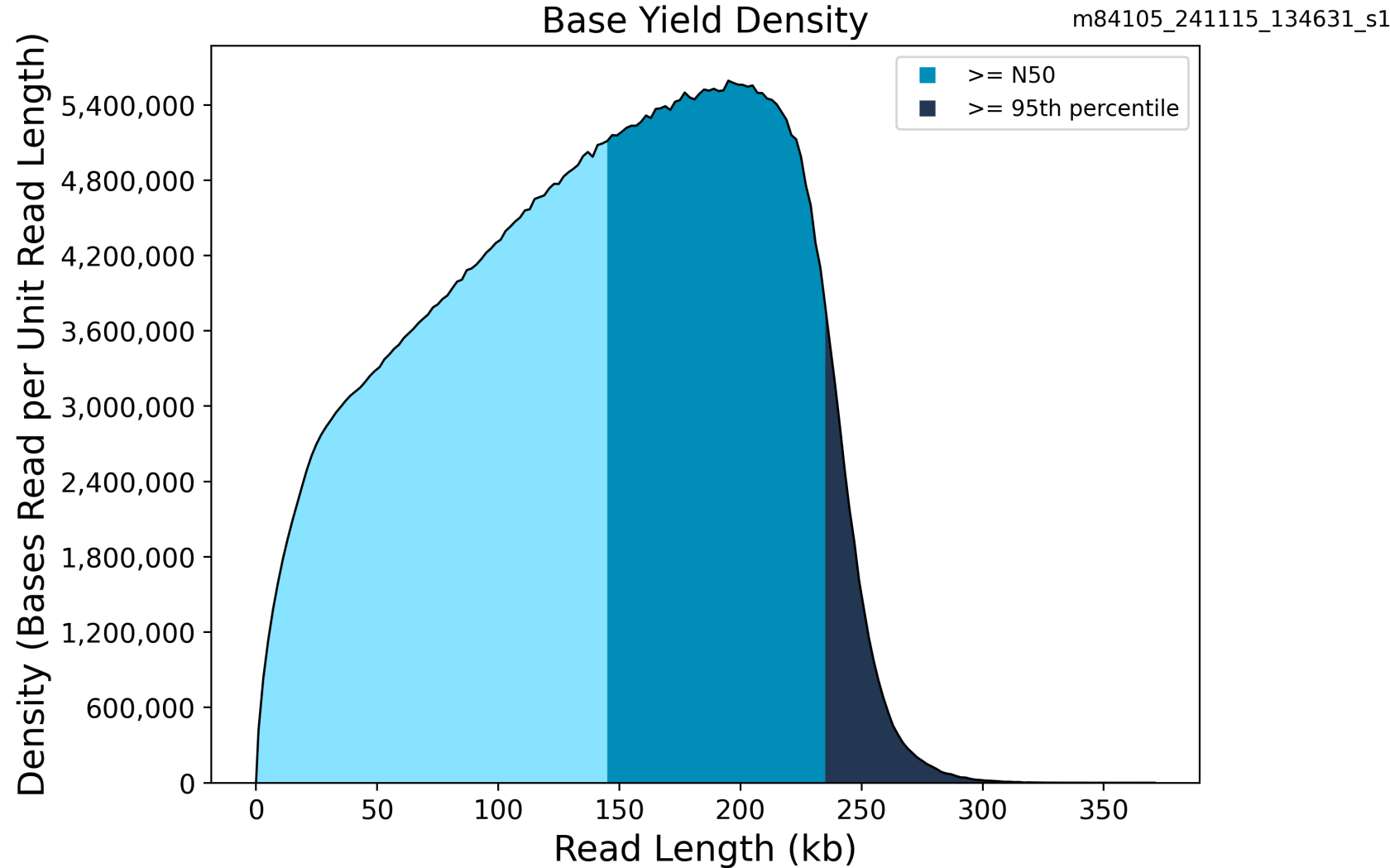
Polymerase Read Length



Longest Subread Length Versus Polymerase Read Length



Base Yield Density



Adapter Report

Summary

Adapter Dimers (0-10bp) %	0
Short Inserts (11-100bp) %	0
Local Base Rate	2.67

CCS Analysis Report

Summary

HiFi reads	5.9 M
HiFi reads yield	86.84 Gb
HiFi reads length (mean)	14.81 kb
HiFi reads length (median, bp)	14,109
HiFi Read Length N50 (bp)	15,118
HiFi Read Quality (median)	Q35
HiFi Read Quality (median)	35
Base Quality Q30 (%)	93.30%
HiFi Number of Passes (mean)	9

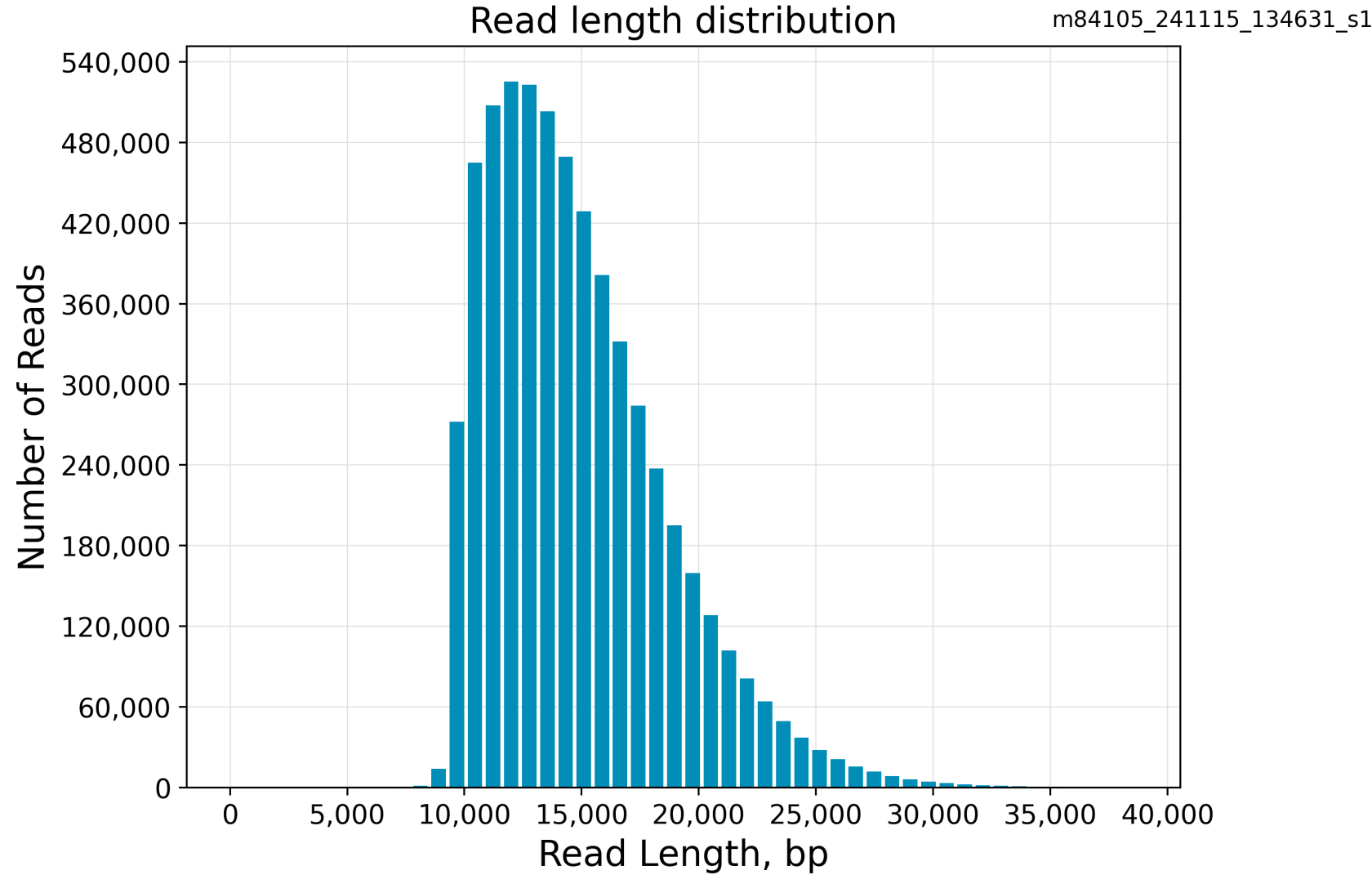
HiFi Read Length Summary

Read Length (kb)	Reads	Reads (%)	Yield (Gb)	Yield (%)
0	5864222	100	86.84 Gb	100
5,000	5863525	100	86.84 Gb	100
10,000	5600873	96	84.30 Gb	97
15,000	2406376	41	44.41 Gb	51
20,000	586597	10	13.36 Gb	15
25,000	95810	2	2.62 Gb	3
30,000	11580	0	0.37 Gb	0
35,000	1129	0	0.04 Gb	0
40,000	126	0	0.01 Gb	0

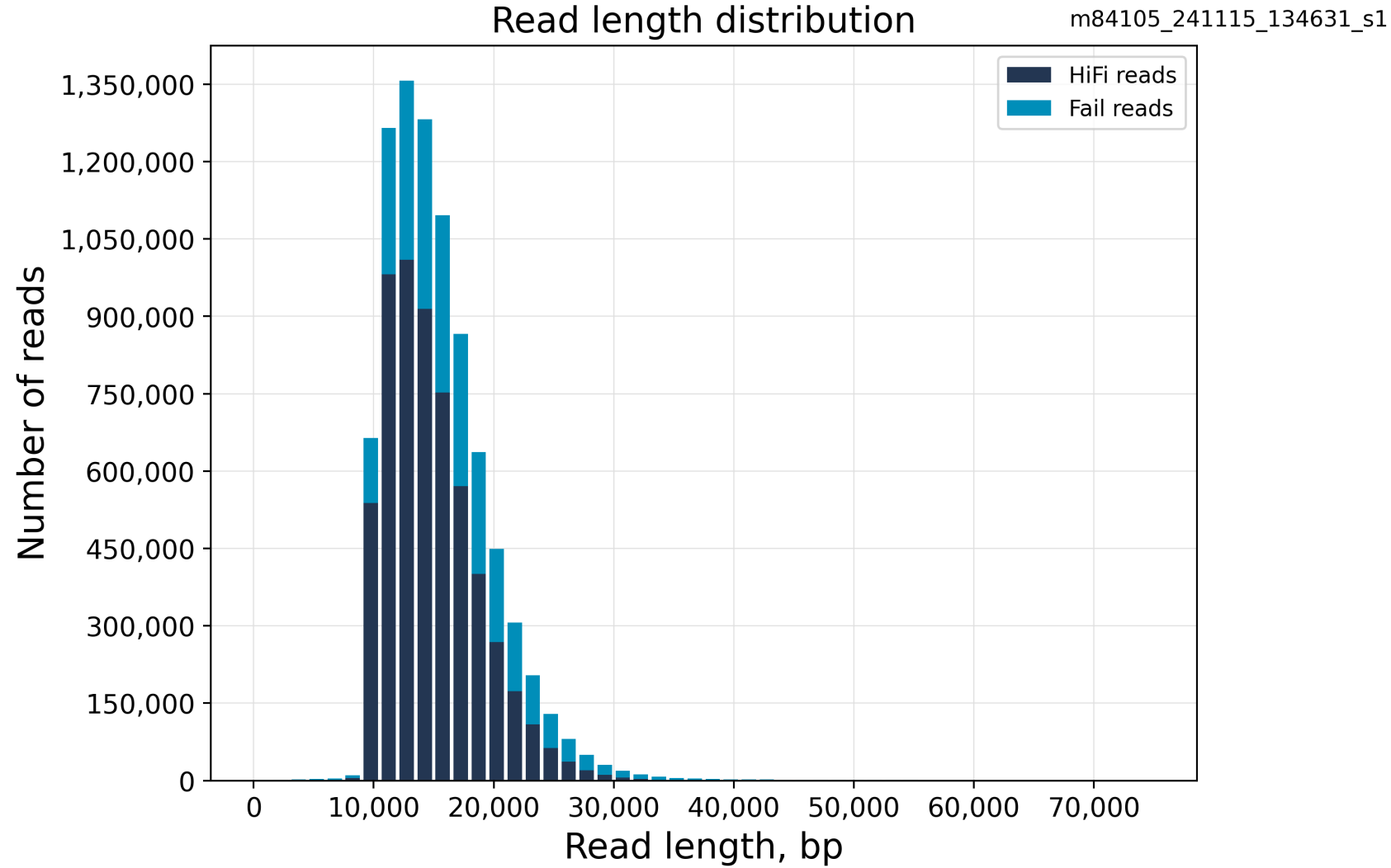
HiFi Read Quality Summary

Read Quality (Phred)	Reads	Reads (%)	Yield (Gb)	Yield (%)
Q20	5864222	100	86.84 Gb	100
Q30	4168651	71	60.01 Gb	69
Q40	1633741	28	21.22 Gb	24
Q50	189901	3	2.14 Gb	2

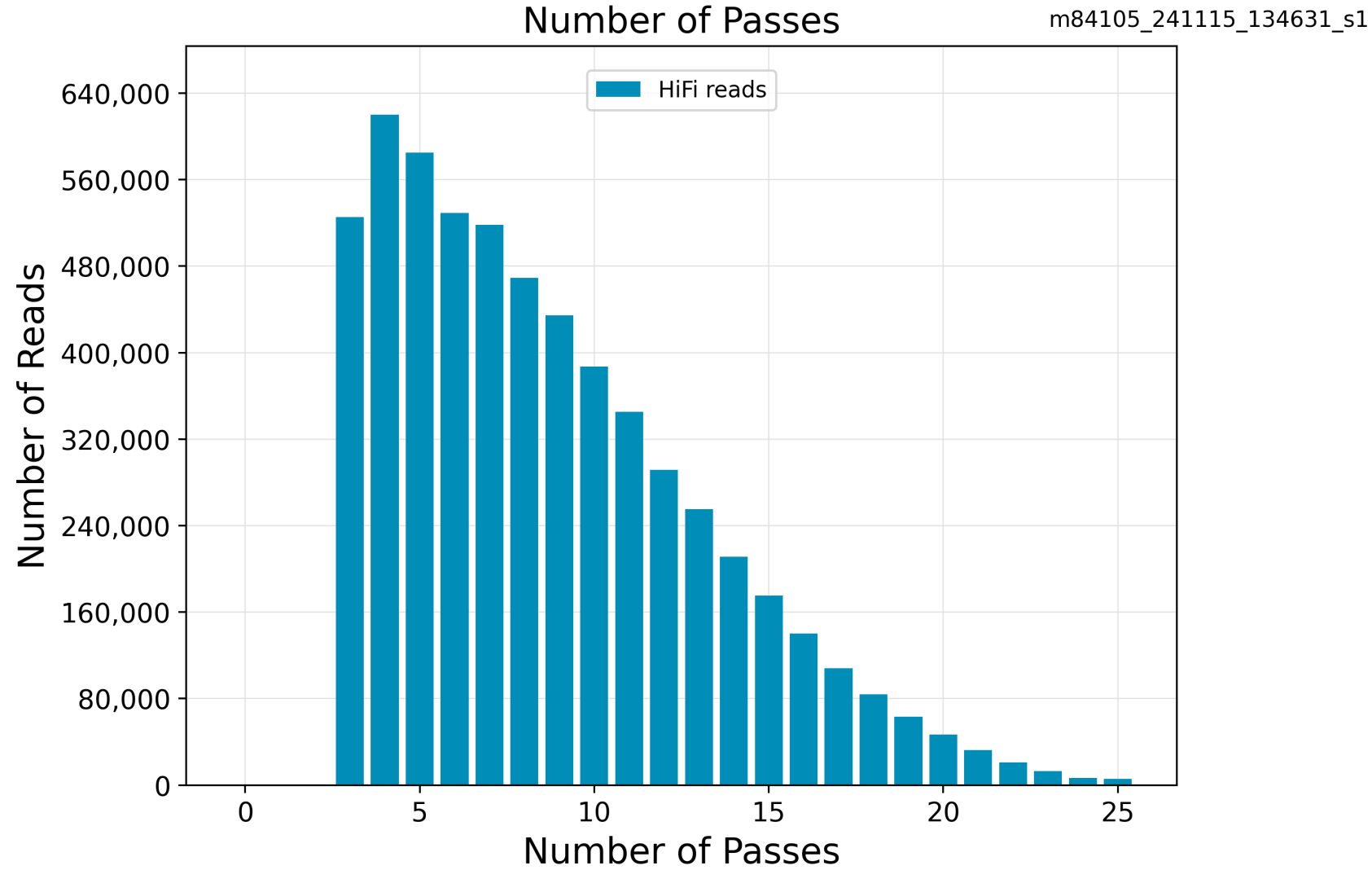
Read length distribution



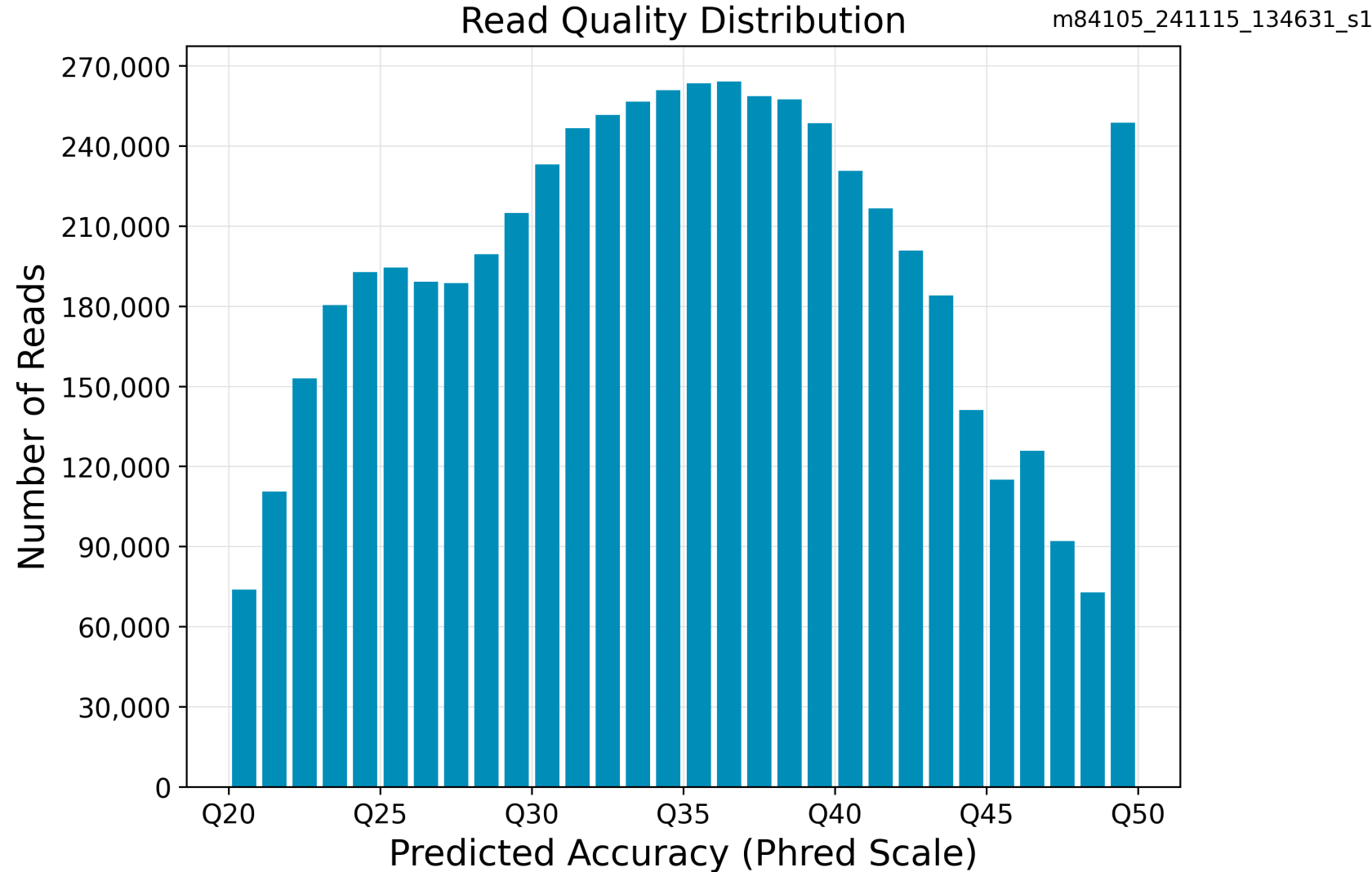
Read length distribution



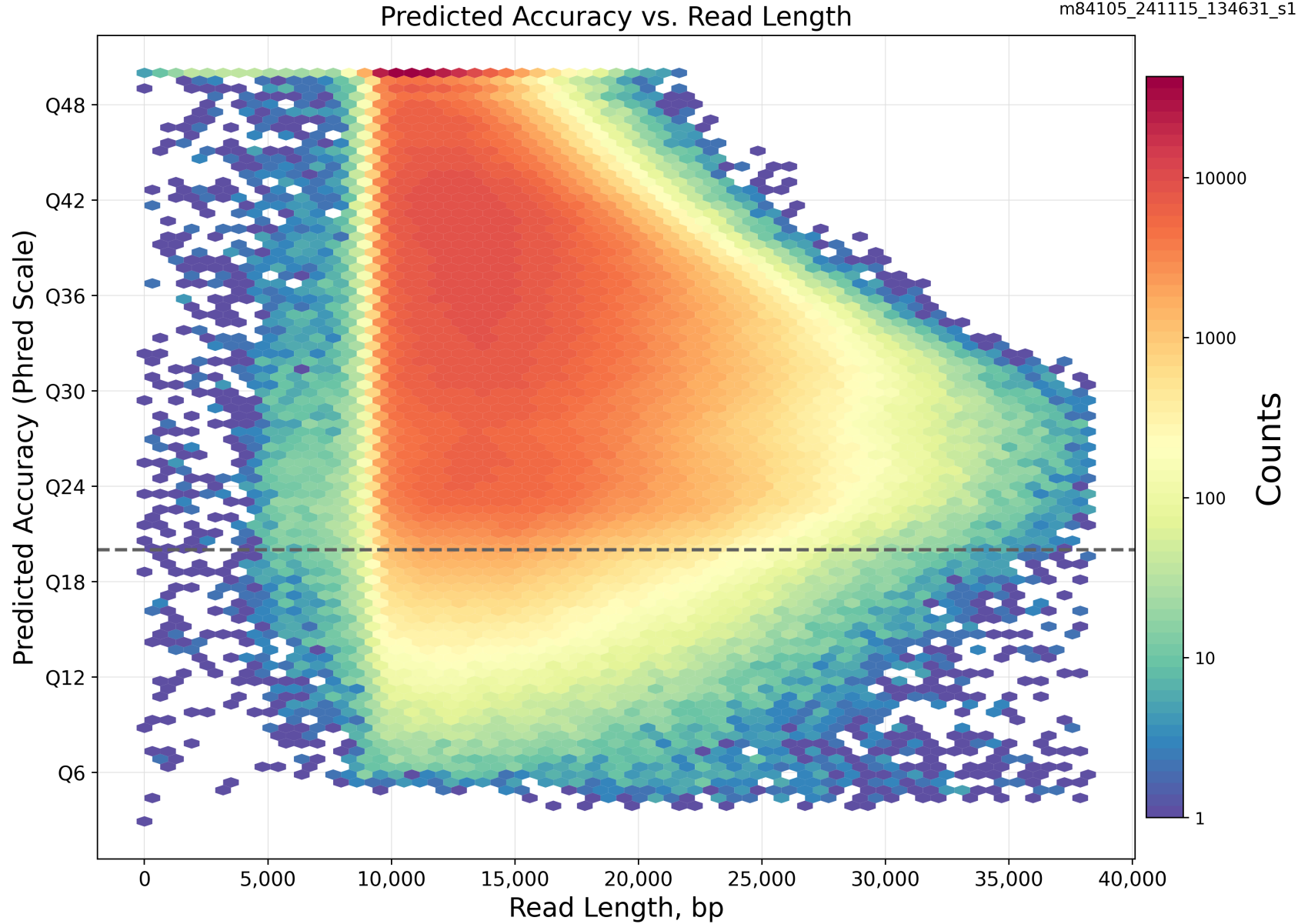
Number of Passes



Read Quality Distribution



Predicted Accuracy vs. Read Length



Loading Report

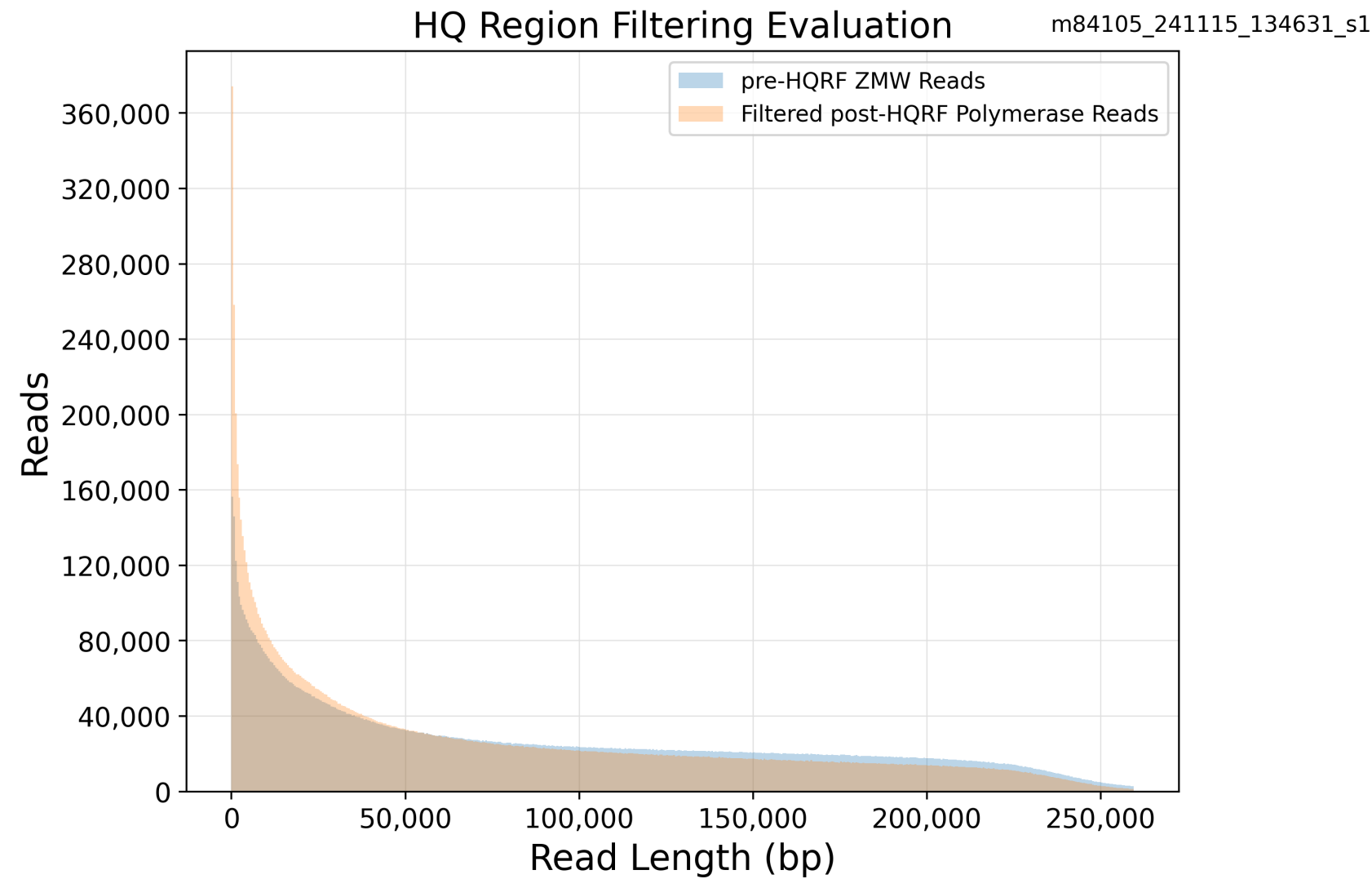
Summary

Productive ZMWs	25,165,824
Productivity 0	11,059,731
Productivity 1	13,983,502
Productivity 2	122,591

Loading Statistics

Collection Context	Total ZMWs	Productivity 0	P0 (%)	Productivity 1	P1 (%)	Productivity 2	P2 (%)	Loading type
m84105_241115_134631_s1	25165824	11059731	43.95	13983502	55.57	122591	0.49	Workflow_Kestrel.py

HQ Region Filtering Evaluation



No Sample Setup found

Instrument run(s)

Run d1c79938-1ff4-4793-92e8-00e68e17a29a

Summary

Name	20241115_Liverpool
Status	COMPLETE
Created	2024-11-15 11:25:11.016
Started	2024-11-15 11:40:08.591
Completed	2024-11-16 22:01:11.298
Context	r84105_20241115_113945
Instrument Name	Revio
Instrument Serial Number	84105
ICS Version	13.1.0.221972
Primary Analysis Version	13.1.0.221972
Chemistry Version	13.1.0.217683

Parent jobs (1)

Job 8764

Summary

Job Type	import-dataset
Name	import-dataset
Comments	Description for job Import PacBio DataSet
Created At	2024-11-17 01:53:51.376
SMRT Link Version	13.1.0.221970

Child jobs (1)

Job 8769

Summary

Job Type	analysis
Pipeline	cromwell.workflows.pb_export_ccs
Name	20241119_Liverpool_Fastq - 20241115_Liverpool_202345_Cell2
Comments	Deferred Job 20241119_Liverpool_Fastq - 20241115_Liverpool_202345_Cell2
Created At	2024-11-19 16:01:48.537
SMRT Link Version	13.1.0.221970

Output files

File name	Size	Type
m84105_241115_134631_s1_fastq.zip	34,161,892,350	zip
m84105_241115_134631_s1_exported.bam	46,685,541,583	bam
Analysis Log	13,543	log
SMRT Link Log	9,545	log

